Translations

1 Consider the point $(8, 6)$.

- **a** Plot the point on the coordinate plane.
- **b** Determine the new coordinates if the point is translated 2 units down.
- **c** Determine the new coordinates if the point from part (b) is translated 6 units to the left.

2 Consider the point $(4, 5)$.

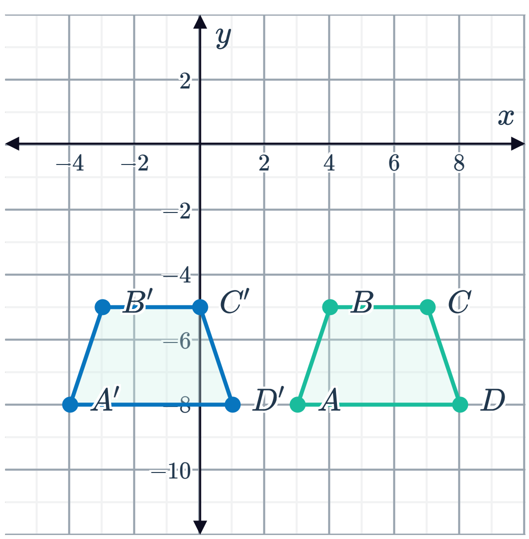
- **a** Plot the point on the coordinate plane.
- **b** Determine the new coordinates if the point is translated 2 units up.
- **c** Determine the new coordinates if the point from part (b) is translated 2 units to the right.

3 Plot the following translations:

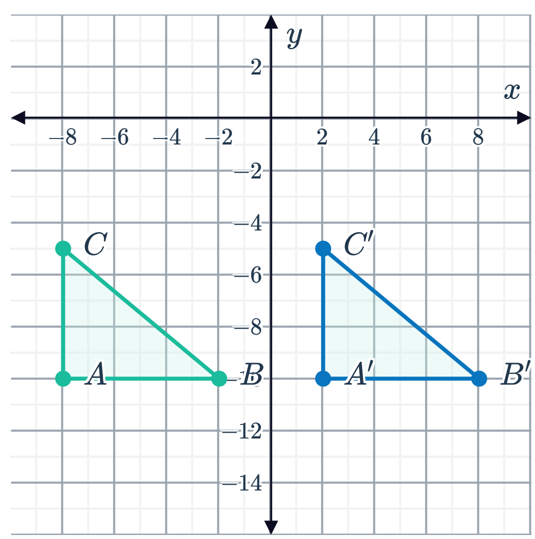
- **a** The point $(-8, 9)$ is translated 18 units to the right.
- **b** The point $(-6, 5)$ is translated 12 units to the right and 7 units down.
- **c** The point $(6, -2)$ is translated 15 units to the left and 6 units up.
- **d** The point $(8, 5)$ is translated 11 units to the left and 9 units down.

4 Describe the transformation that occurs in each graph:

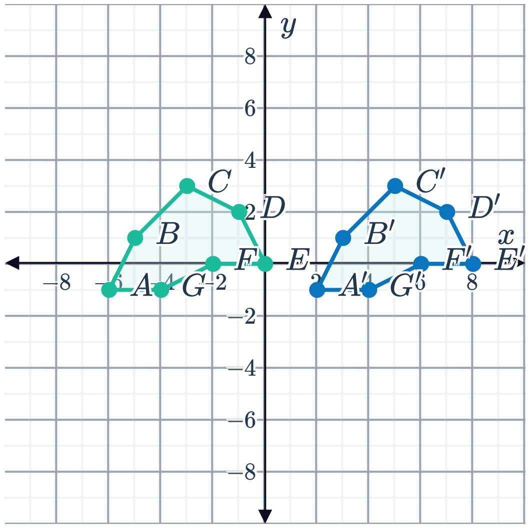
a



b



c


y
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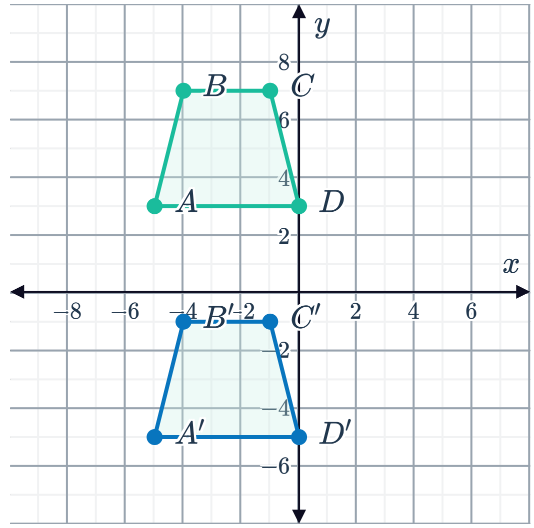
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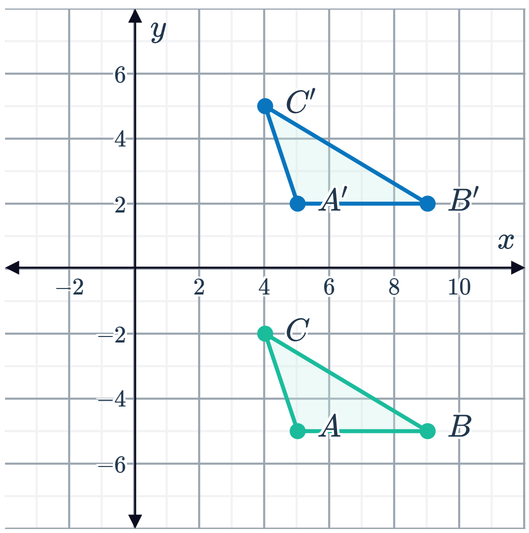
e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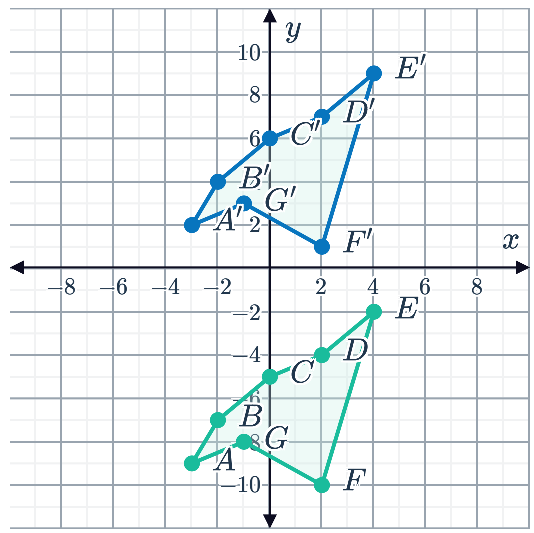
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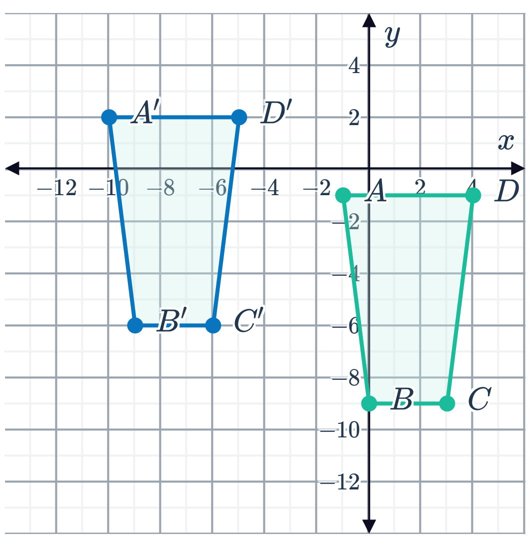
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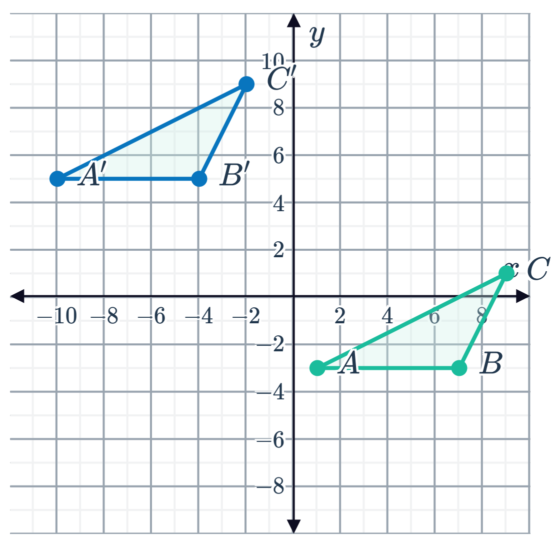
h


y
x
A
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C
A′
B′
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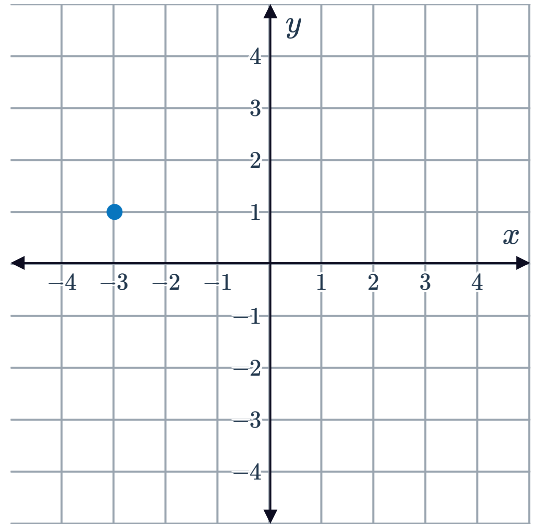
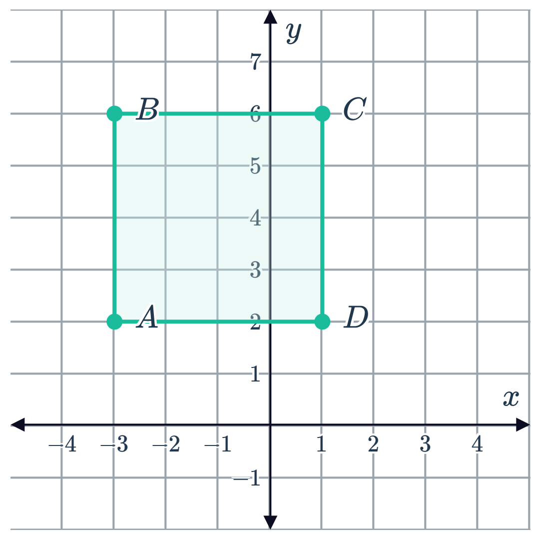
5 The point indicated on the coordinate plane is translated 4 units up. Determine the coordinates of its new position.



6 Quadrilateral $ABCD$ is to be translated 2 units up and 4 units to the left. Determine the coordinates of the vertex A' of the translated rectangle.



7 Determine the coordinates of each of the following translations of the point $A\,(3, -4)$:

a Point A is translated 5 units to the right.

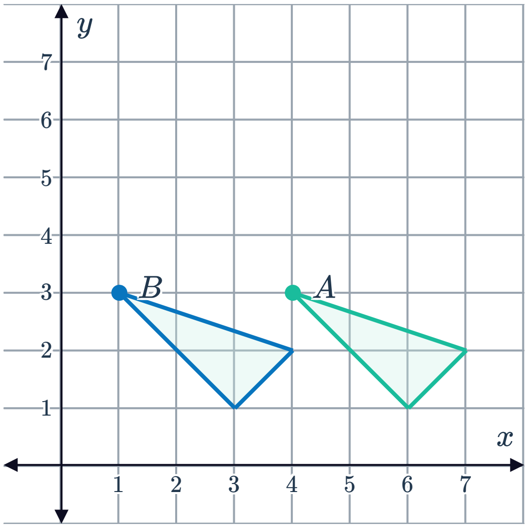
b Point A is translated 5 units to the left.

c Point A is translated 5 units upward.

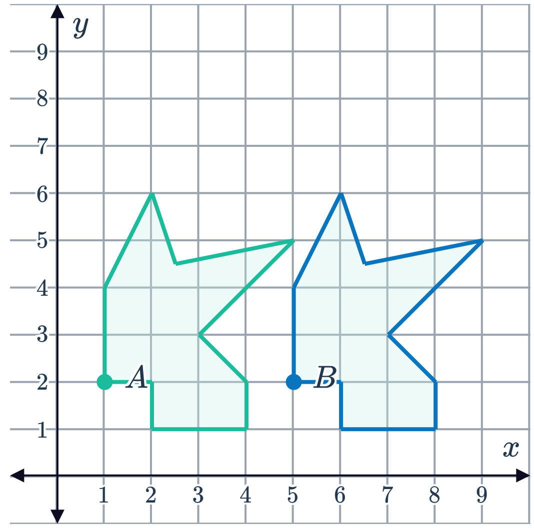
d Point A is translated 5 units downward.

8 Determine the translation from polygon B t[redacted]ygon A.

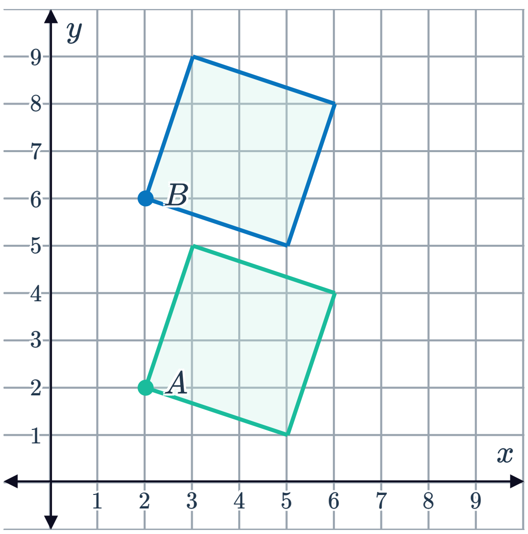
a



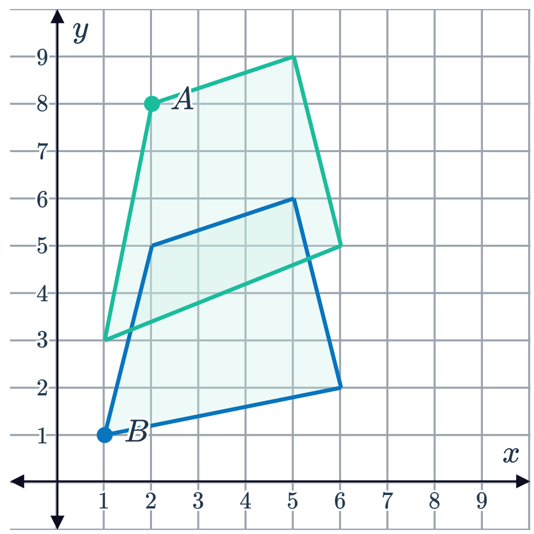
b



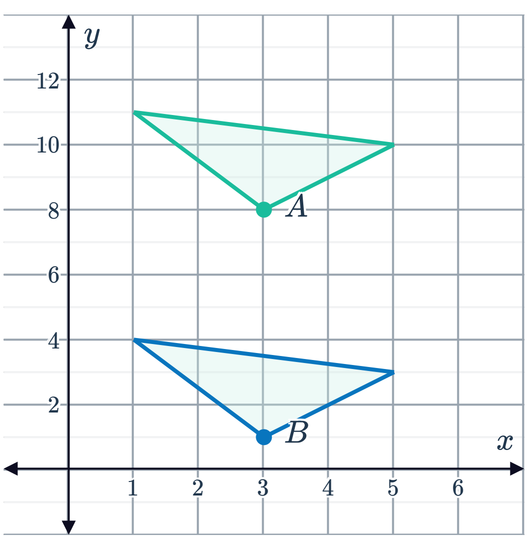
c



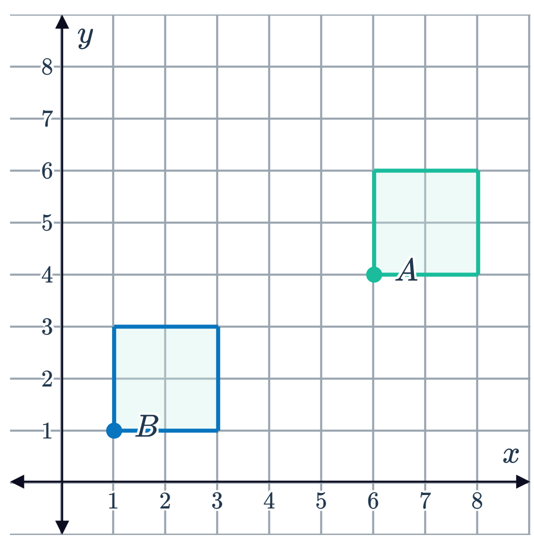
d



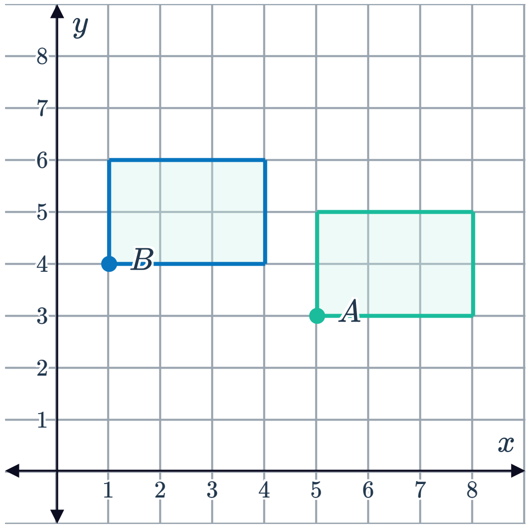
e

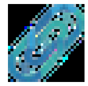


f



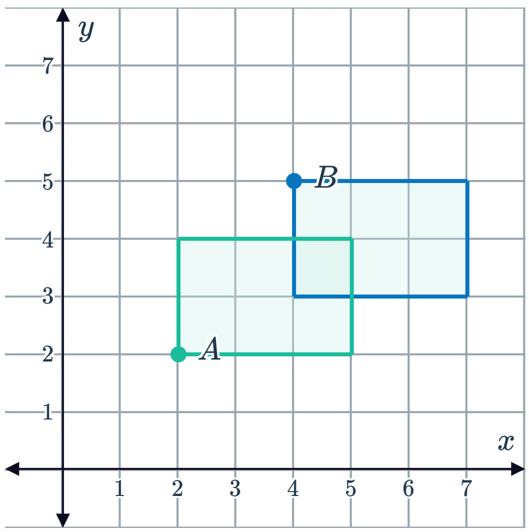
g



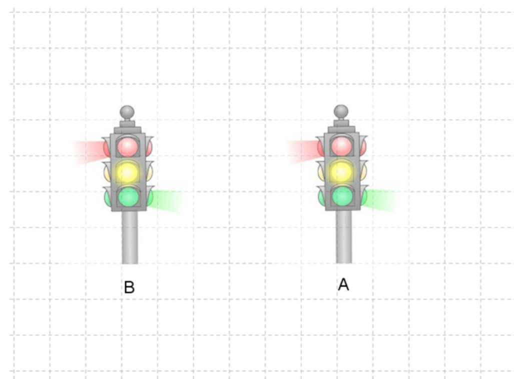


9 Determine the translation from figure A to figure B.

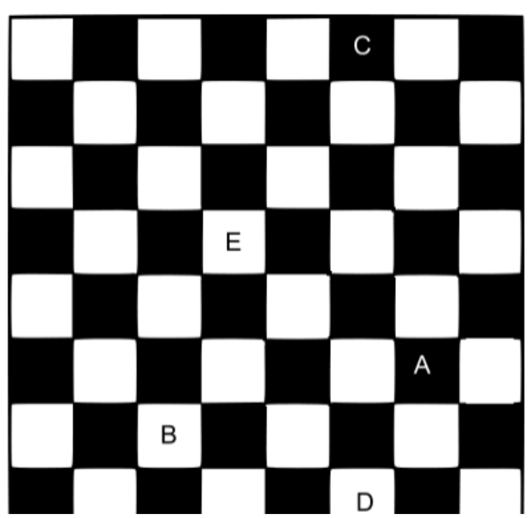
a



b



10 On a chessboard, a knight must move in an L shape. That is, 1 unit in any direction and then 2 units in a perpendicular direction. If a knight is at square E now, what labelled square could it have been two moves ago?

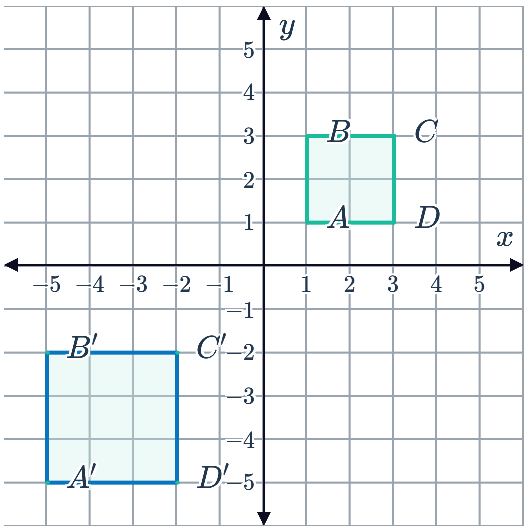


Additional questions

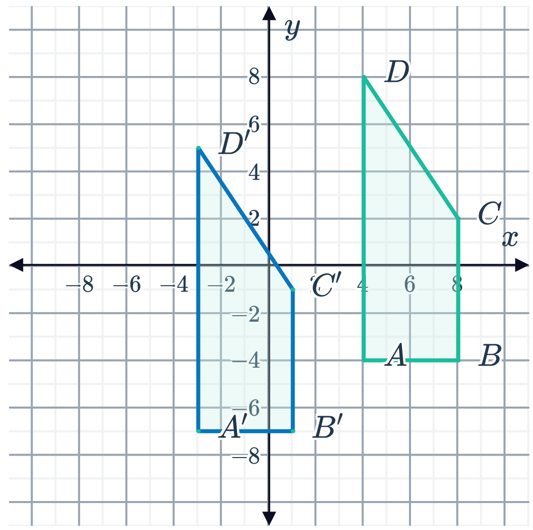
11 Describe the image produced as a result of translating a figure 4 units to the left.

12 Identify whether each graph shows a translation from the preimage to the image. Explain your reasoning.

a



b



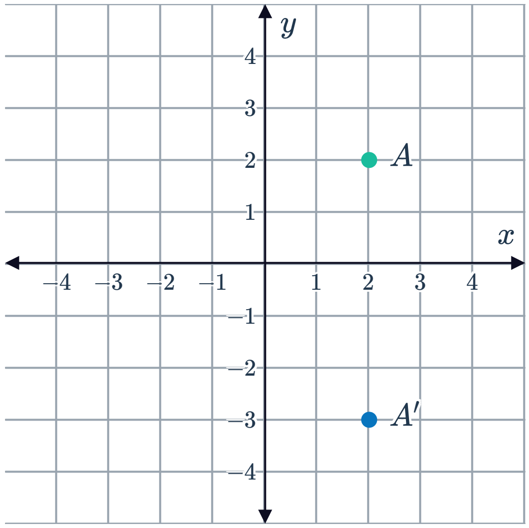
13 Consider the point $(8, 6)$.

a Plot the point on the coordinate plane. Label the point D.

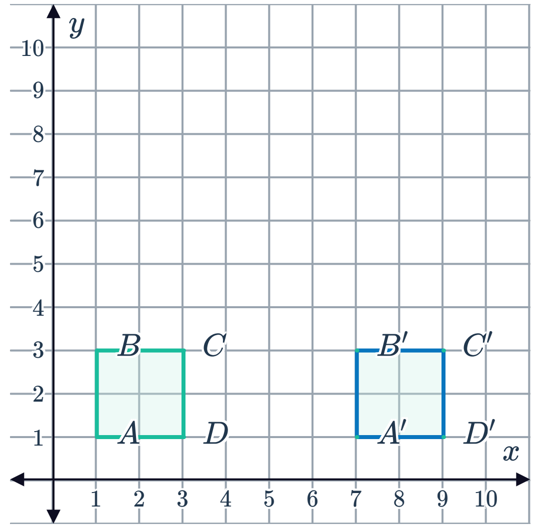
b Translate the point 6 units to the left and 2 units down. Plot the point and label it D'.

14 Describe the transformation that occurs in e[redacted]raph:

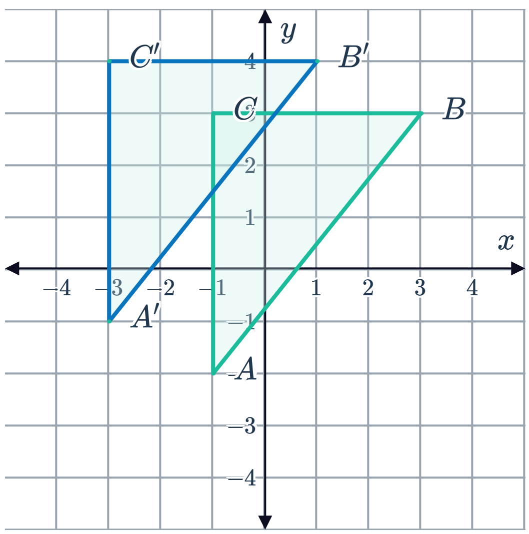
a



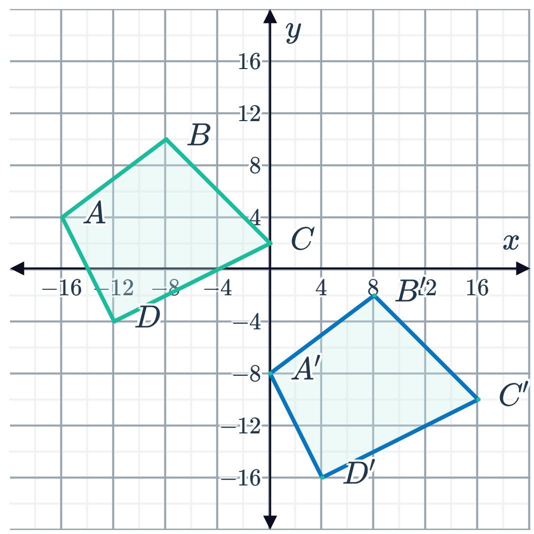
b



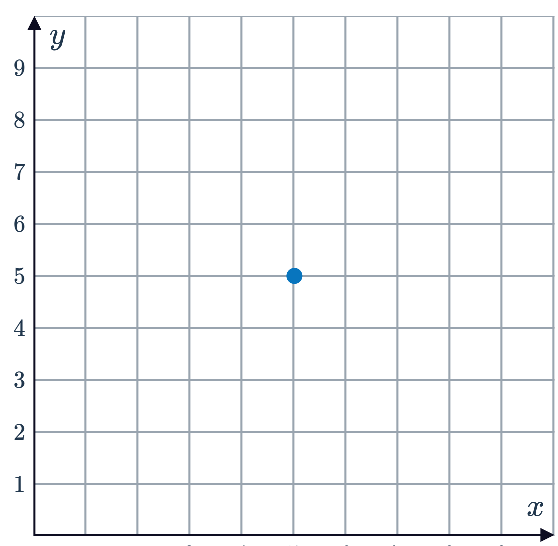
c



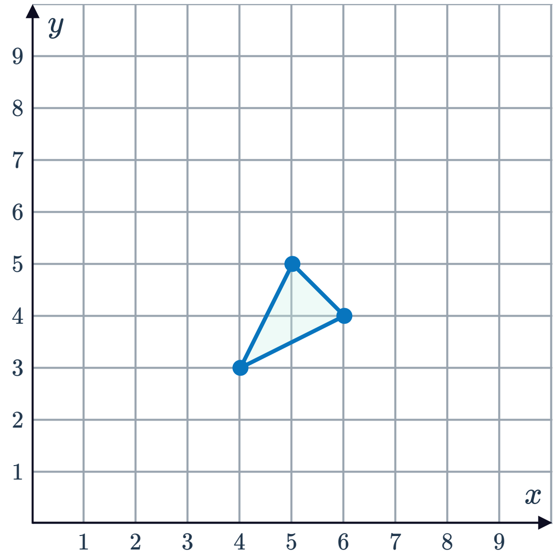
d



15 The point shown is the image produced by the translation: $(x, y) \rightarrow (x-2, y)$.
Plot the starting position of this point before it was translated.

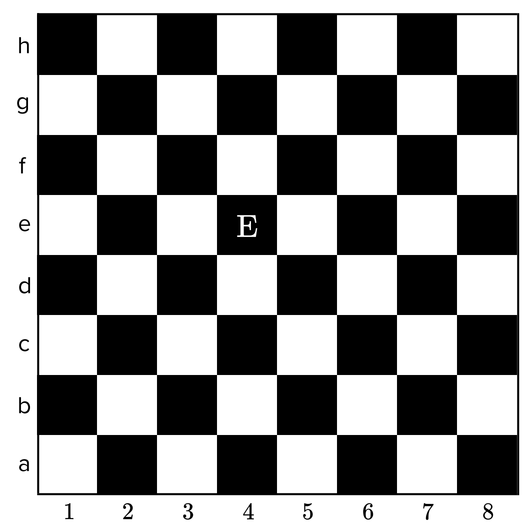


16 The image shown is produced by translating the triangle 3 units to the right and 4 units down.
Plot the starting position of the triangle before it was translated.



17 Point A is translated 3 units down and 4 units to the right, resulting in the image $A'\ (5, -2)$.
Determine the coordinates of point A.

18 In a game of chess, a knight can move around the board in an L -shape. That is, 1 unit in any direction and then 2 units in a perpendicular direction.
If a knight is located at square E now:



a How many possible places could it have been prior to its latest move?

b List all possible places it could have been prior to its latest move.

19 Riccardo has been creating a tower with building blocks.
Is it possible to use the same transformation mapping to place block E or block N at the highest point on the tower? If yes, describe the mapping. If not, explain how you know.